

A Comparison of Phase Recovery Algorithms for Energy Efficient Coherent Receivers

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1 Viterbi & Viterbi

Figure 1 shows the operation of the Viterbi-Viterbi algorithm. A block of symbols $[z[k - N], z[k - N + 1], \dots, z[k], \dots, z[k + N]]$ is raised to the 4th power to remove data dependency and convolved with a maximum-likelihood (ML) filter of length $L = 2N + 1$ to compute $\hat{\theta}_{\text{ML}}[k]$. A phase unwrapper is applied to reflect the unbounded Wiener phase noise, then the unwrapped phase correction is applied to $z[k]$. Here $M = 4$ for QAM.

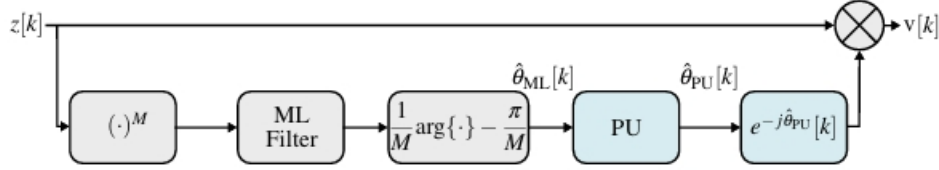


Figure 1: Viterbi & Viterbi algorithm block diagram [1]

An energy efficient hardware implementation of this algorithm avoids multipliers where possible. The ML filter is known beforehand so the multiplications can be replaced with fixed additions and shifts. Computing the argument of the filter output and the phase correction can be implemented with CORDIC. The unwrapped phase is computed as

$$\hat{\theta}_{\text{PU}}[k] = \hat{\theta}_{\text{ML}}[k] + n[k] \frac{\pi}{2},$$

where $n[k] \in \{-1, 0, 1\}$ is calculated as

$$n[k] = \left\lfloor \frac{1}{2} + \frac{\hat{\theta}_{\text{PU}}[k - 1] - \hat{\theta}_{\text{ML}}[k]}{\pi/2} \right\rfloor.$$

The floor operation can be implemented efficiently with 2 comparators and division by $\pi/2$ can be implemented as multiplication by $2/\pi$ with fixed shifts and additions as before. However the 4th power operation requires complex multiplications which increase the energy cost. Assuming Gauss's algorithm for complex multiplication, the total cost of this algorithm per symbol in terms of real multiplications (RM), real additions (RA), and bit-shifts (BS) for n -bit fixed point precision with N_C CORDIC iterations is shown in Table 1. Note that the number of shifts and adds for multiplication by a fixed value

would be h and $h - 1$, where h is the Hamming weight, but here it is approximated as n for simplicity. The cost of a comparator operation is assumed to be the same as a real addition.

Operation	RM	RA	BS
4th Power	$9L$	$15L$	0
ML Filter	0	$2(n - 1)L$	nL
ML Phase	0	$3N_C$	$2N_C$
Phase Unwrapper	0	$5 + 2(n - 1)$	$2n$
Phase Correction	0	$3N_C$	$2N_C$
Total	$9L$	$2nL + 13L + 6N_C + 2n + 3$	$nL + 4N_C + 2n$

Table 1: Energy cost of Viterbi-Viterbi algorithm per symbol

2 Blind Phase Search

The Blind Phase Search algorithm is an alternative feed-forward algorithm that can be implemented without multipliers. Figure 2 shows its operation. Producing the test rotations θ_b can be done very efficiently with CORDIC by reusing intermediate values, resulting in only $\sum_{b=1}^{\log_2 B} 2^{b+1} = 4B - 4$ shift and add operations for B rotations [2]. For a QAM constellation $\mathcal{S}$, the decision circuit is implemented with $2(\sqrt{|\mathcal{S}|} - 1)$ comparators. Forming the squared distances $|d_b[k]|^2$ appears to require a multiplier but the low degree of precision necessary means a look-up table (LUT) can be used [2]. Additive noise is filtered from the estimate by summing $L = 2N + 1$ squared distances. Finding the minimum $m_b[k]$ can be done with $B - 1$ comparators.

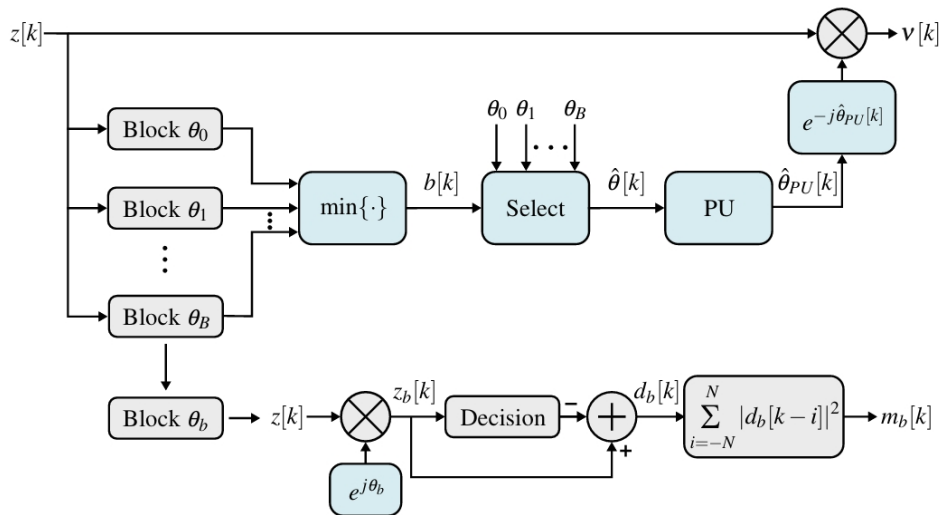


Figure 2: Blind Phase Search algorithm block diagram [1]

The cost of this algorithm per symbol is shown in Table 2. For small constellations, the LUT will be small so accesses will be assumed to have negligible cost along with any MUX operations to select the correct θ_b .

Operation	RM	RA	BS
Test Rotations	0	$4B - 4$	$4B - 4$
Forming $d_b[k]$	0	$2(\sqrt{ \mathcal{S} } - 1) + 2$	0
Forming $m_b[k]$	0	L	0
Minimum Search	0	$B - 1$	0
Phase Unwrapper	0	$5 + 2(n - 1)$	$2n$
Phase Correction	0	$3N_C$	$2N_C$
Total	0	$5B + 2\sqrt{ \mathcal{S} } + L + 2n + 3N_C - 2$	$4B + 2n + 2N_C - 4$

Table 2: Energy cost of Blind Phase Search algorithm per symbol

3 Computational Cost

As each stage of CORDIC applies a left bit-shift, the accuracy achieved by N_C iterations is approximately N_C bits. Thus setting sensible values of $N_C = n = 8$, $L = 5$, $|\mathcal{S}| = 16$, and $B = 32$, the total cost of each algorithm is shown in Table 3.

Algorithm	RM	RA	BS
Viterbi-Viterbi	45	212	88
Decision-Directed	0	71	92

Table 3: Energy cost comparison of Viterbi & Viterbi and Decision-Directed algorithms per symbol

It is clear the the Blind Phase Search algorithm is more energy efficient. This is due to the absence of multipliers, a simpler additive noise filter, and efficient reuse of intermediate values for test rotations.

4 Phase Estimation Frequency

Both algorithms above produce a phase noise estimate for every symbol. The phase noise is slowly varying so only estimating every K symbols and applying the same correction to the intermediate symbols may be sufficient. This would reduce the cost of both algorithms by a factor of K at the cost of increased phase noise variance. Figure 3 shows the effect of K on the phase noise variance for both algorithms. Both algorithms use pilots to avoid cycle slips [3] and exhibit almost identical performance.

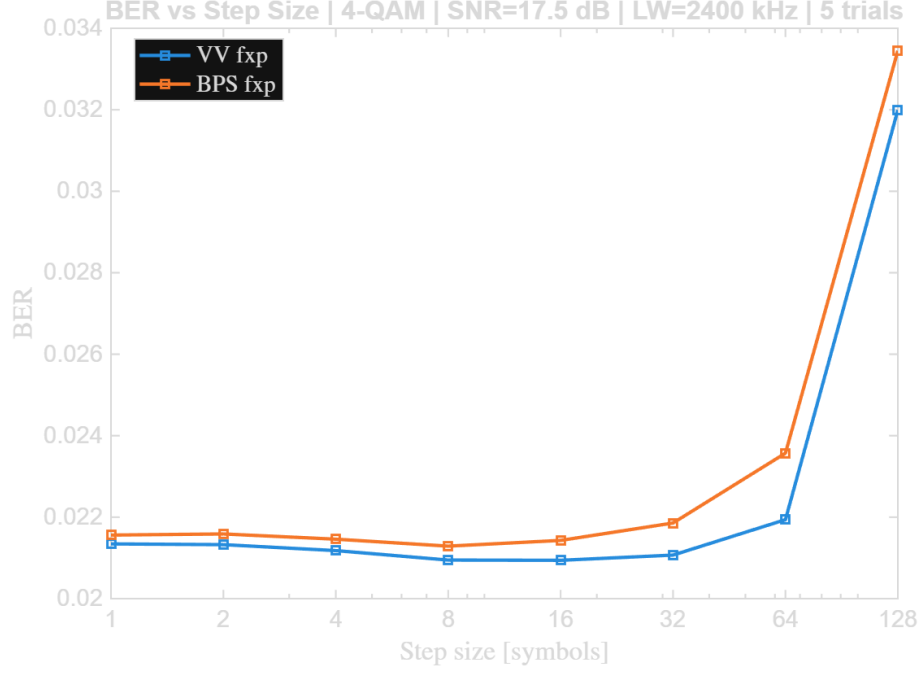


Figure 3: Phase noise variance vs phase estimation frequency K

5 Filter Length

Blind Phase Search averages over several minimum distances to filter out noise. It can be seen from Table 2 that the cost of this filter is linear in L so it should be made as small as possible while still giving acceptable performance. Figure 4 shows the effect of L on the phase noise variance.

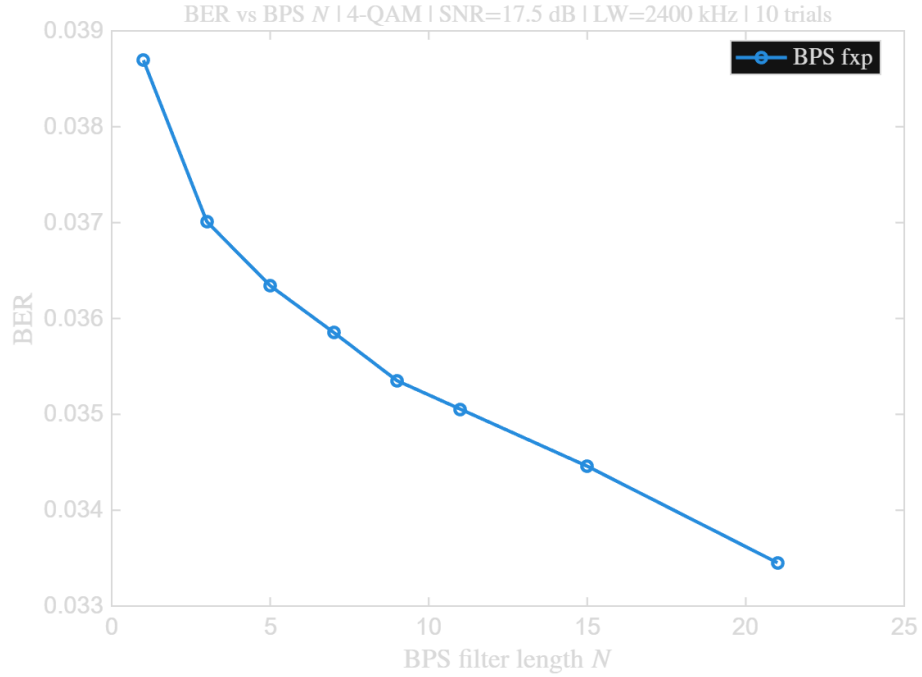


Figure 4: Phase noise variance vs filter length L

References

- [1] D. A. de Arruda Mello and F. A. Barbosa, *Digital Coherent Optical Systems: Architecture and Algorithms*. Optical Networks, Cham, Switzerland: Springer Nature, 2021.
- [2] T. Pfau, S. Hoffmann, and R. Noe, “Hardware-efficient coherent digital receiver concept with feedforward carrier recovery for m-qam constellations,” *Journal of Lightwave Technology*, vol. 27, no. 8, pp. 989–999, 2009.
- [3] H. Cheng, Y. Li, F. Zhang, J. Wu, J. Lu, G. Zhang, J. Xu, and J. Lin, “Pilot-symbols-aided cycle slip mitigation for dp-16qam optical communication systems,” *Optics Express*, vol. 21, no. 19, pp. 22166–22172, 2013.